

LiveTradeBench Seeking Real-World Alpha with Large Language Models

來源：<https://arxiv.org/abs/2511.03628>

作者：Haofei Yu, Fenghai Li, Jiaxuan You

發表時間：5 Nov 2025

引用次數: 0



Introduction

- 研究目的: 建立一個能在真實市場中即時運作的交易環境, 以評估 LLM 在動態、不確定環境下的決策能力與金融行為。

Table 1: Comparison of LiveTradeBench with existing trading benchmarks. We compare our work with others through four dimensions: (1) *sequential decision* for whether its current trading actions rely on the previous actions; (2) *portfolio management* for whether its task is multi-asset portfolio management; (3) *live trading* for whether the evaluation belongs to backtest with historical market data or live test with real-time streaming data; (4) *multi-market evaluation* for whether it includes markets beyond the stock market.

Benchmark	Sequential Decision	Portfolio Management	Live Trading	Multi-market Evaluation
FinQA (Chen et al., 2021)	✗	✗	✗	✗
ConvFinQA (Chen et al., 2022)	✗	✗	✗	✗
FLUE (Shah et al., 2022)	✗	✗	✗	✗
FinEval (Zhang et al., 2023)	✗	✗	✗	✗
BizFinBench (Bigeard et al., 2025)	✗	✗	✗	✗
FinAgentBench (Bigeard et al., 2025)	✗	✗	✗	✗
FinSearchComp (Hu et al., 2025)	✓	✗	✗	✗
INVESTORBENCH (Li et al., 2024)	✓	✗	✗	✗
StockBench (Chen et al., 2025)	✓	✓	✗	✗
DeepFund (Li et al., 2025b)	✓	✓	✓	✗
LiveTradeBench (ours)	✓	✓	✓	✓

Building Live Trading Environment for Portfolio Management

- Market Selection

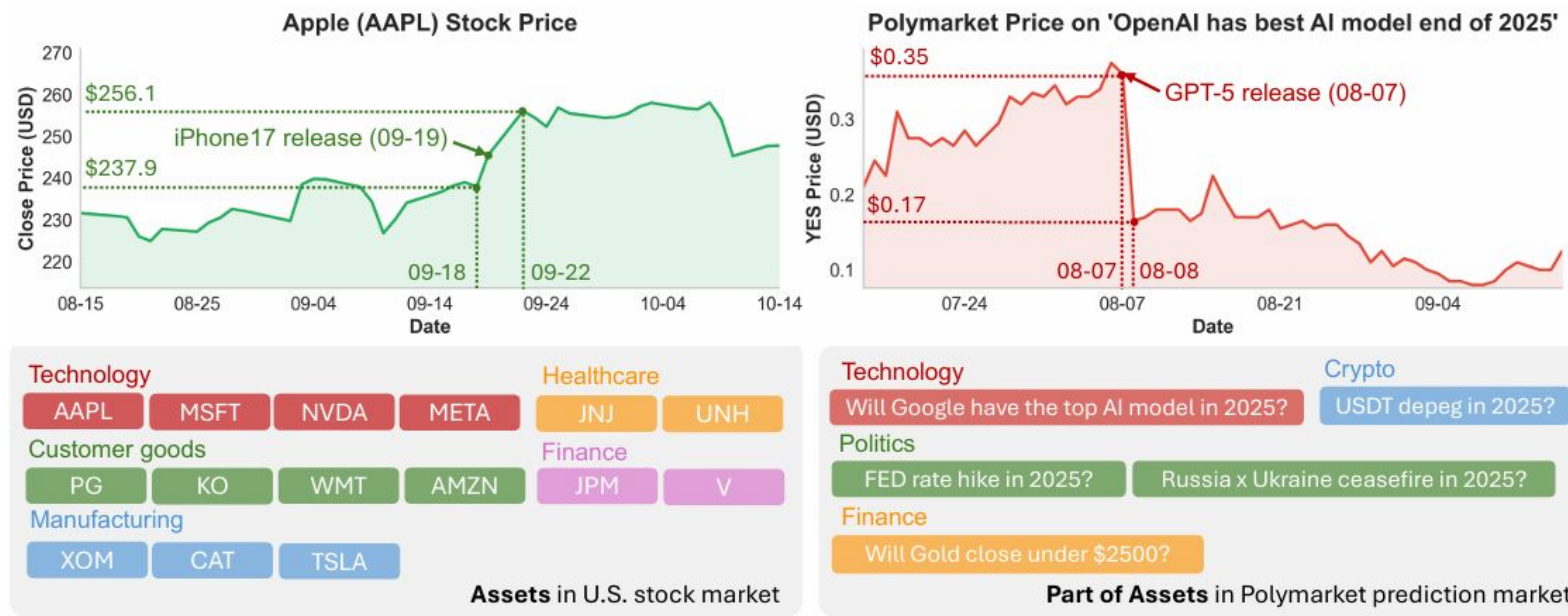


Fig. 1: Market selection in LiveTradeBench. The top panels show AAPL in the U.S. stock market (left) and the contract “OpenAI has the best AI model by the end of 2025” in the Polymarket prediction market (right). In prediction markets, the price directly reflects the probability of a given outcome. Both markets respond to news and historical price trends, but Polymarket exhibits sharper fluctuations, faster reactions, and higher sensitivity to external signals. The bottom panels display representative assets across various domains, including technology, finance, cryptocurrency, manufacturing, and politics.

Building Live Trading Environment for Portfolio Management

- Observation Space
- Action Space

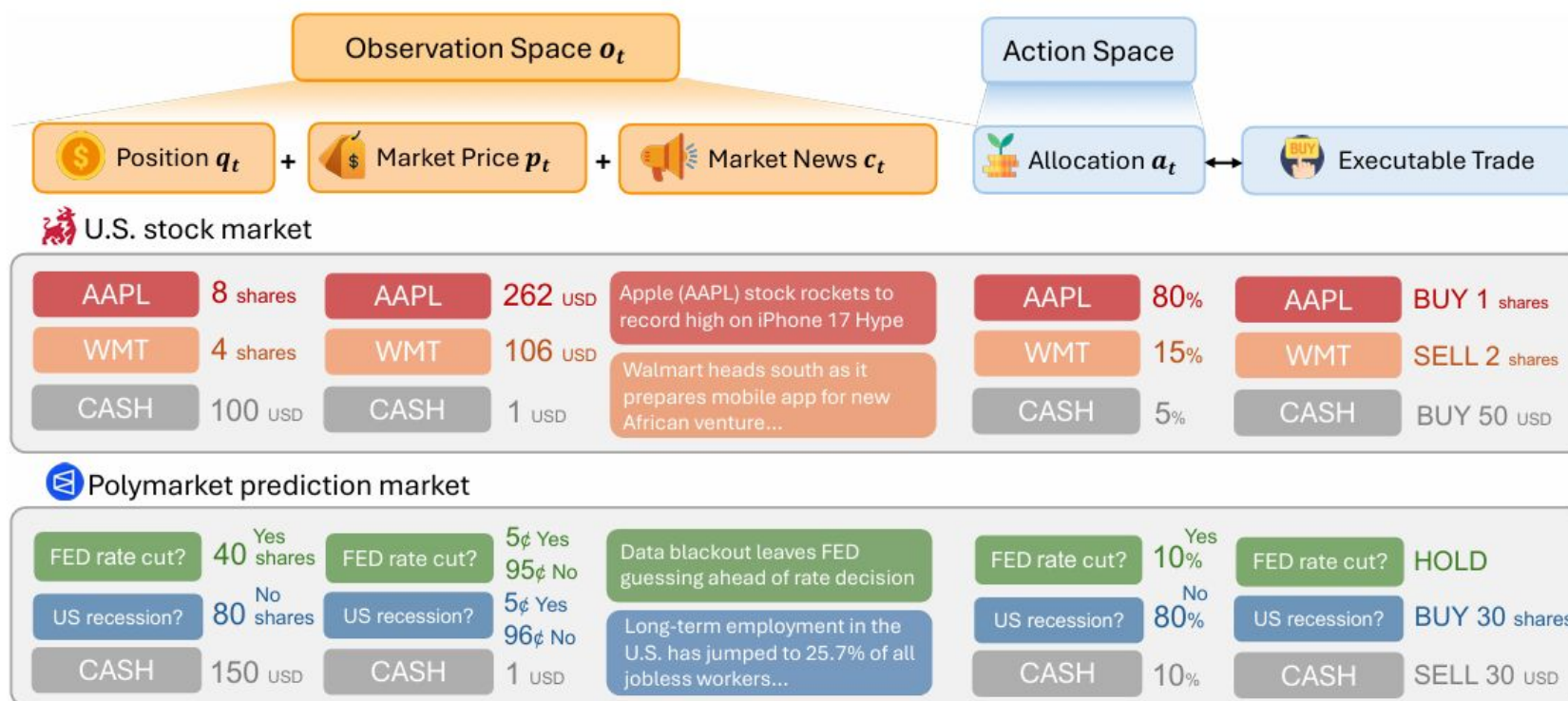


Fig. 2: Observation and action space for LiveTradeBench. We illustrate examples from both the U.S. stock market and the Polymarket prediction market to demonstrate the observation and action spaces. The observation space consists of three components: the agent's position, market prices, and relevant news context. The action space represents the portfolio allocation decisions generated by the agent, which can be directly translated into executable trading actions.

Designing LLM-based Agents for Portfolio Management

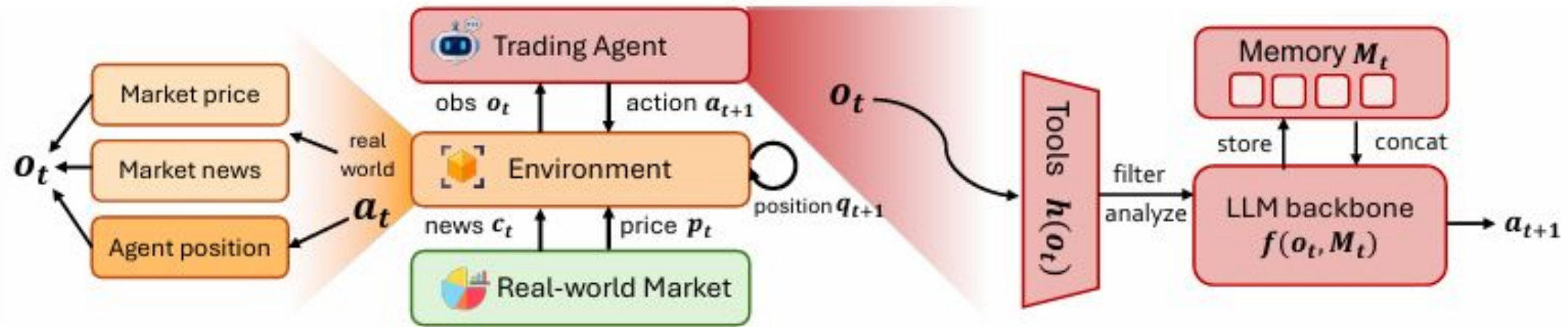


Fig. 3: Agent and environment framework in LiveTradeBench. The left side illustrates the simulated environment, which continuously retrieves real-world market prices and news, updating its internal state accordingly. It also adjusts the agent's portfolio position based on the executed actions. The right side depicts the portfolio-management agent, equipped with analytical tools to process observations from the environment. The agent maintains a memory of past observations, enabling adaptive and context-aware decision-making.

Evaluating LLM-based Agents under Live Test

- 時間: 2025 年 8 月 18 日到 2025 年 10 月 24 日(總共 50 個交易日)
- Backbone LLMs for Evaluation
 - Claude (Sonnet、Opus 4/4.1)
 - Grok (Grok-3、Grok-4)
 - Qwen (72B, 235B Instruct, 235B Thinking)
 - Llama (Llama3.3、Llama4 Scout、Llama4 Maverick)
 - GPT (GPT-4o、GPT-4.1、GPT-5、GPT-o3)
 - Kimi (K2)
 - DeepSeek (V3.1、R1)

Evaluating LLM-based Agents under Live Test

- Trading Metrics for Evaluation

- Cumulative Return $(CR = \frac{v_T - v_0}{v_0})$

- Sharpe Ratio $(SR = \frac{\bar{r} - r_f}{\sigma})$

- Maximum Drawdown $(MDD = \max_{t \in [1, T]} \frac{\max_{i \in [1, t]} v_i - v_t}{\max_{i \in [1, t]} v_i})$

- Win Rate $(WR = \frac{1}{T-1} \sum_{t=2}^T \mathbb{I}(r_t > 0))$

- Volatility $(\sigma = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2})$

Evaluation Results

Table 2: Comparison of trading performance across U.S. stock and Polymarket prediction markets. We use five key metrics: cumulative return (CR), Sharpe ratio (SR), maximum drawdown (MDD), win rate (WR), and volatility (σ). The highest value in each column is highlighted in bold.

Model	U.S. Stock Market					Polymarket Prediction Market				
	CR $\uparrow$	SR $\uparrow$	MDD $\downarrow$	WR $\uparrow$	$\sigma\downarrow$	CR $\uparrow$	SR $\uparrow$	MDD $\downarrow$	WR $\uparrow$	$\sigma\downarrow$
Claude-Sonnet-3.7	3.63	1.45	2.65	59.18	10.25	20.54	2.38	10.65	51.02	44.64
Claude-Sonnet-4	2.45	0.72	3.23	53.06	12.86	-40.32	-2.40	51.10	38.78	92.16
Claude-Opus-4	3.93	1.72	2.11	63.27	9.45	-2.04	0.09	13.67	46.94	56.38
Claude-Opus-4.1	3.73	1.51	1.84	63.27	10.17	-25.69	-3.02	30.53	48.98	46.81
Grok-3	3.22	1.26	2.13	65.31	10.17	-8.35	-0.55	18.80	53.06	54.35
Grok-4	4.30	1.75	1.92	59.18	10.43	7.38	1.01	13.04	46.94	46.92
Qwen2.5-72B-Instruct	5.15	2.18	2.22	65.31	10.24	1.63	0.43	7.46	59.18	30.36
Qwen3-235B-A22B-Instruct	3.52	1.32	2.04	61.22	10.89	-54.24	-2.97	54.24	40.82	112.37
Qwen3-235B-A22B-Thinking	1.78	0.60	2.32	59.18	9.28	-57.62	-1.81	72.10	38.78	166.92
Llama3.3-70B-Instruct-Turbo	2.72	0.88	3.54	61.22	11.95	1.58	0.40	19.41	57.14	38.15
Llama4-Scout	4.65	1.99	2.62	59.18	9.98	-16.05	-1.18	24.80	51.02	60.37
Llama4-Maverick	4.46	1.65	2.45	53.06	11.59	-18.31	-1.92	28.63	34.69	48.21
GPT-4o	3.55	1.39	2.43	55.10	10.38	-30.96	-3.26	35.31	30.61	53.75
GPT-4.1	6.25	2.64	1.92	65.31	10.51	-33.69	-1.74	37.98	40.82	95.27
GPT-5	5.31	2.19	2.53	65.31	10.60	-23.96	-0.49	38.92	44.90	130.37
GPT-o3	6.04	2.57	2.27	61.22	10.41	-54.84	-3.68	60.99	40.82	97.27
Gemini-2.5-Flash	2.10	0.72	3.10	55.10	11.25	-22.40	-0.82	42.35	38.78	115.42
Gemini-2.5-Pro	1.95	0.61	2.85	50.00	10.98	-35.15	-1.65	49.80	34.69	101.87
Kimi-K2-Instruct	3.07	1.15	3.32	53.06	10.53	-53.44	-5.26	54.74	28.57	69.41
DeepSeek-V3.1	2.46	0.86	2.45	59.18	10.61	-4.68	-0.07	22.43	48.98	64.74
DeepSeek-R1	2.10	0.78	2.20	61.22	9.11	-13.19	0.14	44.16	42.86	143.25

Analysis and Discussion

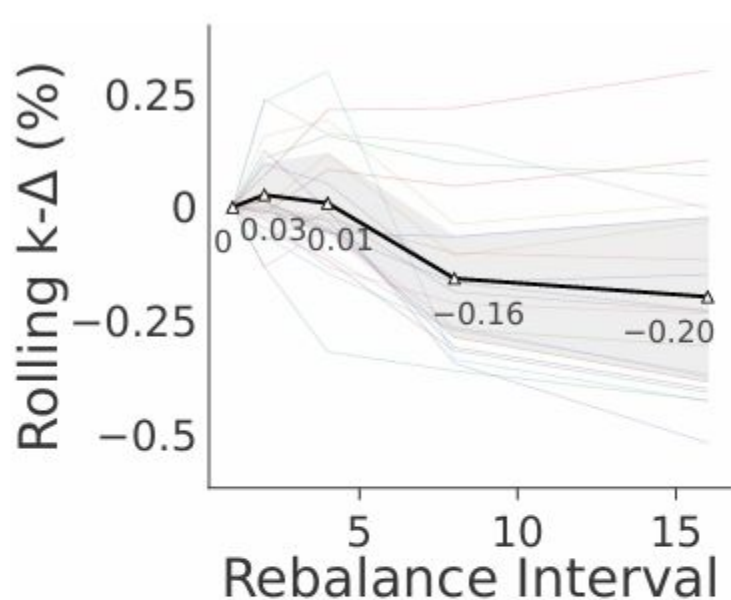


Fig. 5: Rolling k -delta analysis on U.S. stocks. We evaluate rebalance intervals $k \in \{1, 2, 4, 8, 16\}$. The black line denotes the mean performance across 21 models, and the shaded gray region indicates the 25–75% confidence interval.

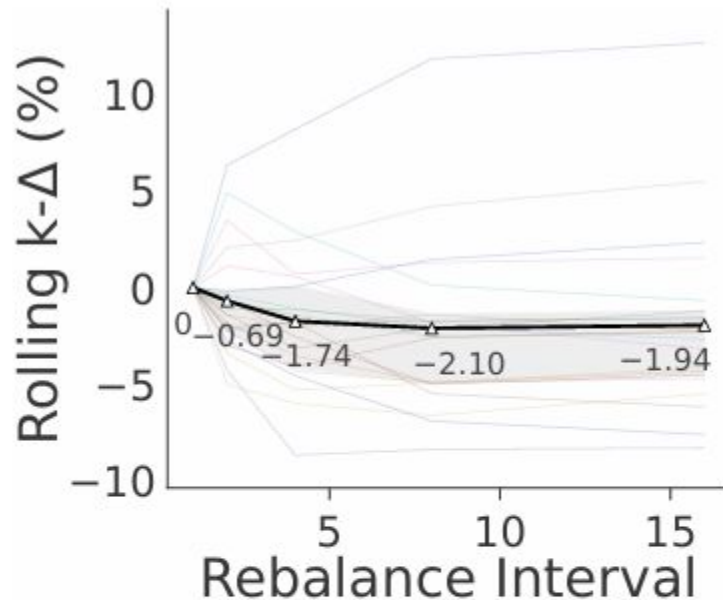


Fig. 6: Rolling k -delta analysis on Polymarket. We also evaluate rebalance intervals $k \in \{1, 2, 4, 8, 16\}$. The black line denotes the mean performance across 21 models, and the shaded gray region indicates the 25–75% confidence interval.



Fig. 7: Decision-making rationale analysis. Each bar indicates the proportion of reasoning traces that reference position, price, or news information. A single reasoning trace may include multiple information sources.

Case Study

0.2 10% of data to 9



Fig. 8: Case study for U.S. stock markets. (Left) The average cash ratio across 21 models over 50 trading days. (Right) A zoomed-in view of the sharp drawdown on October 10, during which portfolios exhibited a sudden increase in cash holdings. We visualize the market condition (price change) and present the reasoning traces of the best-performing model on October 10 (Gemini-2.5-Pro) and one of the worst-performing models on October 10 (DeepSeek-V3.1) for comparison.



Fig. 9: Case study in Polymarket prediction markets. (Left) The average holding ratios of “Yes” and “No” position ratios in the market “Russia × Ukraine ceasefire in 2025?” across 21 models. (Right) A zoomed-in view of two abrupt shifts (October 13 and October 17), along with the corresponding news events and the reasoning traces of Grok-3 explaining these allocation decisions.

TradingAgents: Multi-Agents LLM Financial Trading Framework

來源：<https://arxiv.org/abs/2412.20138>

作者：Yijia Xiao, Edward Sun , Di Luo , Wei Wang

發表時間：28 Dec 2024

引用次數: 89

2025.12.16 報告人：張凱鈞

Introduction

- 解決問題:現有金融 LLM 多是單一代理或缺乏真實公司式的合作結構
- 論文貢獻: TradingAgents 建立了一個模仿交易公司的多角色框架
 - 模擬真實交易團隊決策
 - 基本面分析師、新聞/情緒分析師、技術分析師、交易員
 - 加強與LLM對話能力
 - 透過結構化的方式, 使每個代理的狀態都被明確定義
 - 實驗與驗證

Role Specialization

- 設計了七種不同的代理角色：
 - 基本面分析師 (Fundamentals Analyst)
 - 情緒分析師 (Sentiment Analyst)
 - 新聞分析師 (News Analyst)
 - 技術分析師 (Technical Analyst)
 - 研究員 (Researcher)
 - 交易員 (Trader)
 - 風險管理者 (Risk Manager)

Role Specialization

Why multi-role

A. Analyst Team

組成角色：基本面分析師、情緒分析師、新聞分析師、技術分析師

One agent
with multi-
function

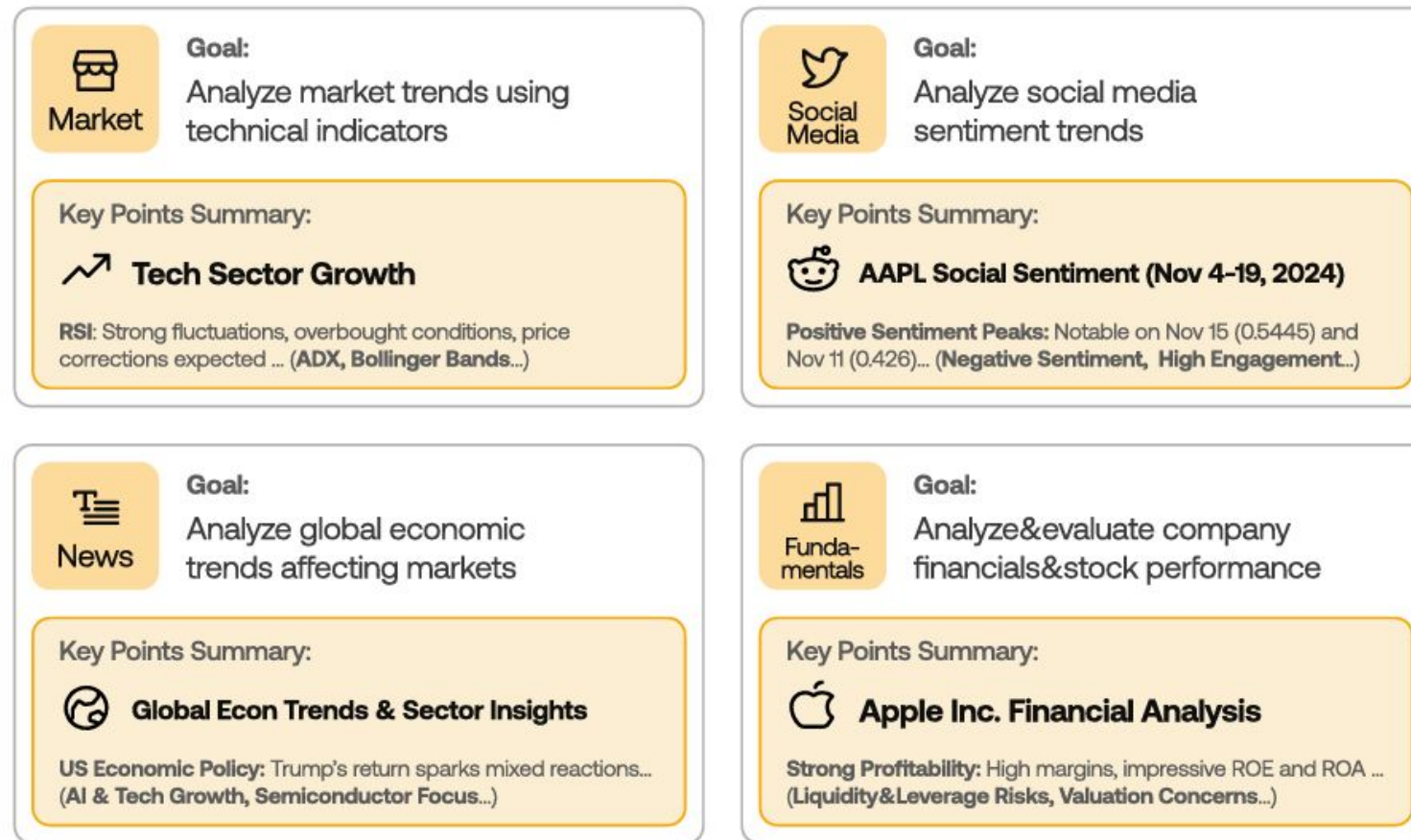


Fig. 2: TradingAgents Analyst Team

Role Specialization

B. Researcher Team

組成角色: 研究員

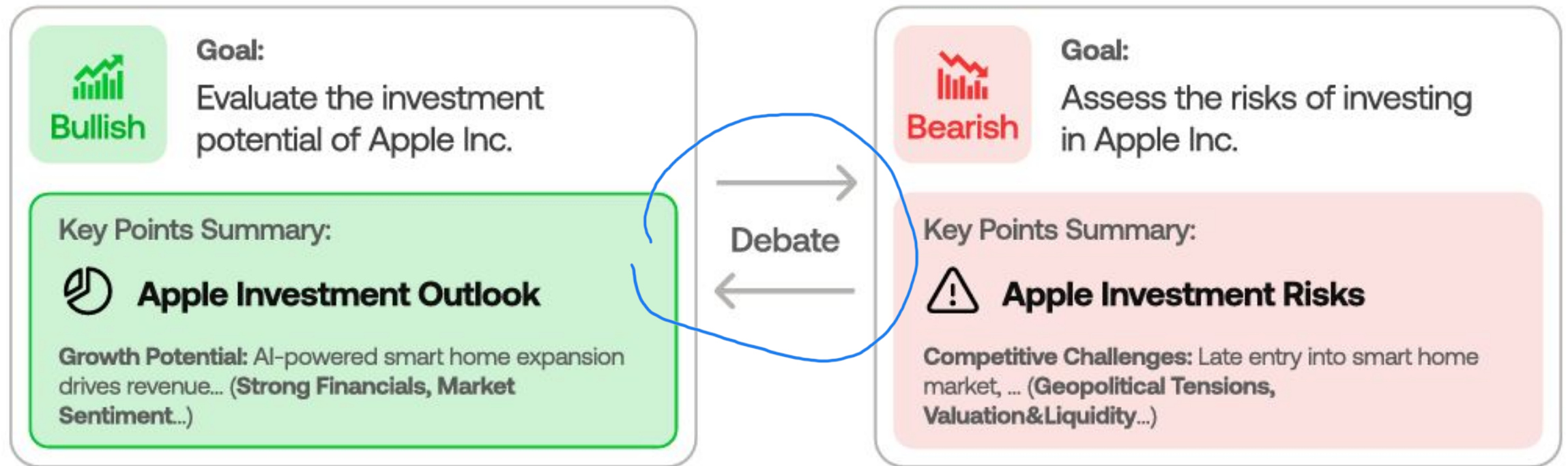


Fig. 3: TradingAgents Researcher Team: Bullish Perspectives and Bearish Perspectives

Role Specialization

C. Trader Agents

組成角色: 交易員

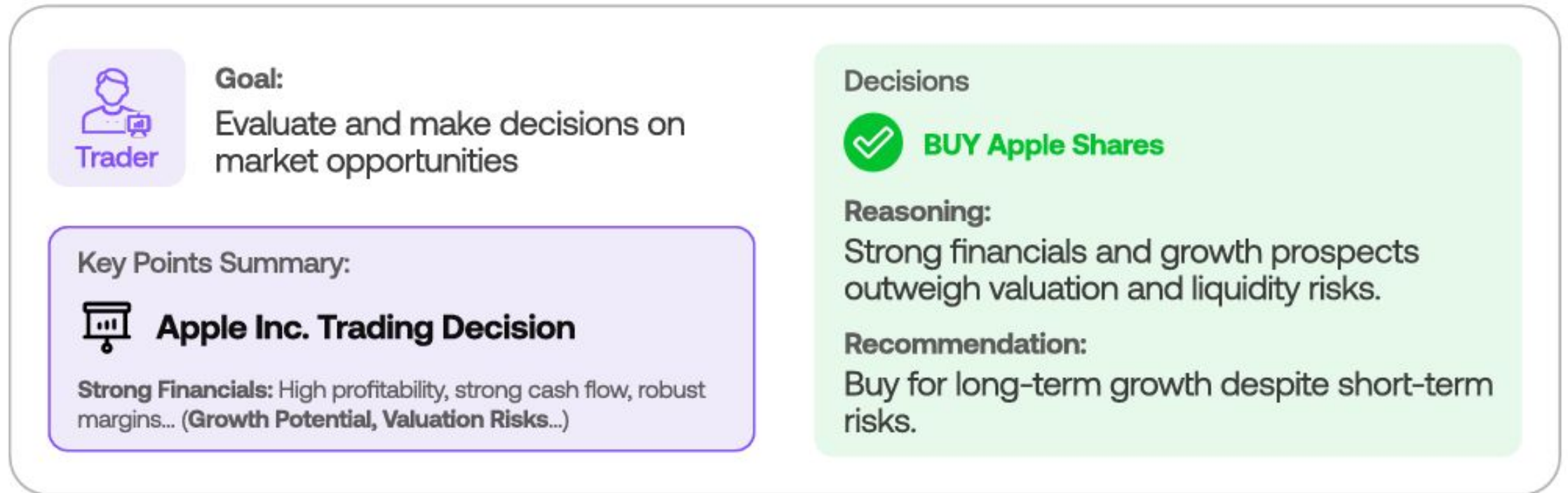


Fig. 4: TradingAgents's Trader Decision-Making Process

Role Specialization

D. Risk Management Team

組成角色: Risk Management

who final judge

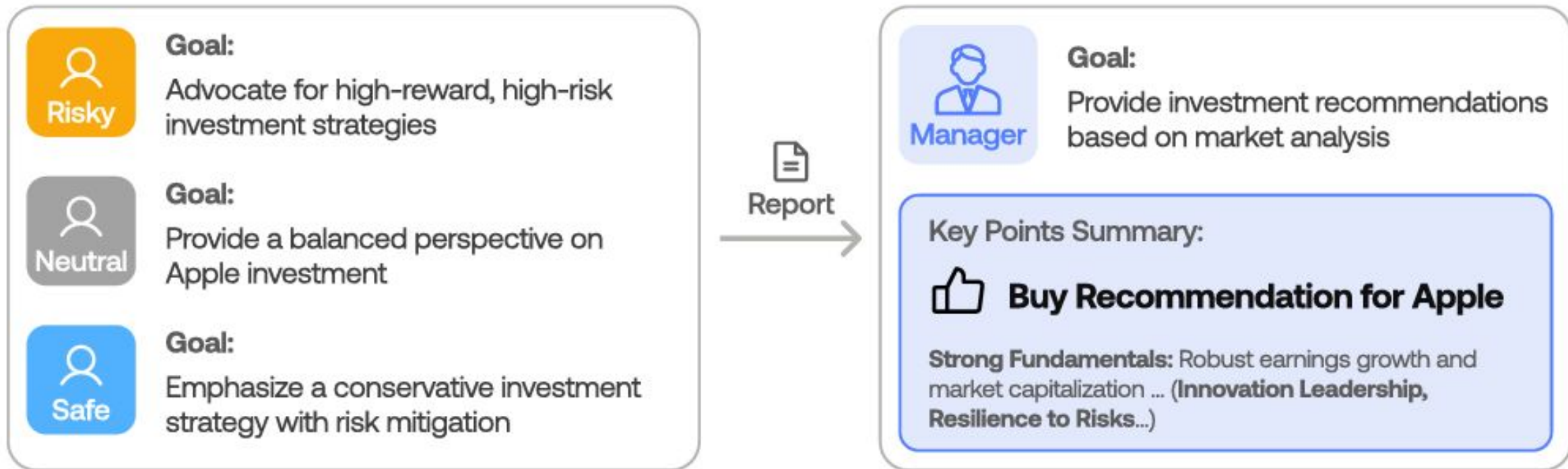


Fig. 5: TradingAgents Risk Management Team and Fund Manager Approval Workflow

Agent Workflow

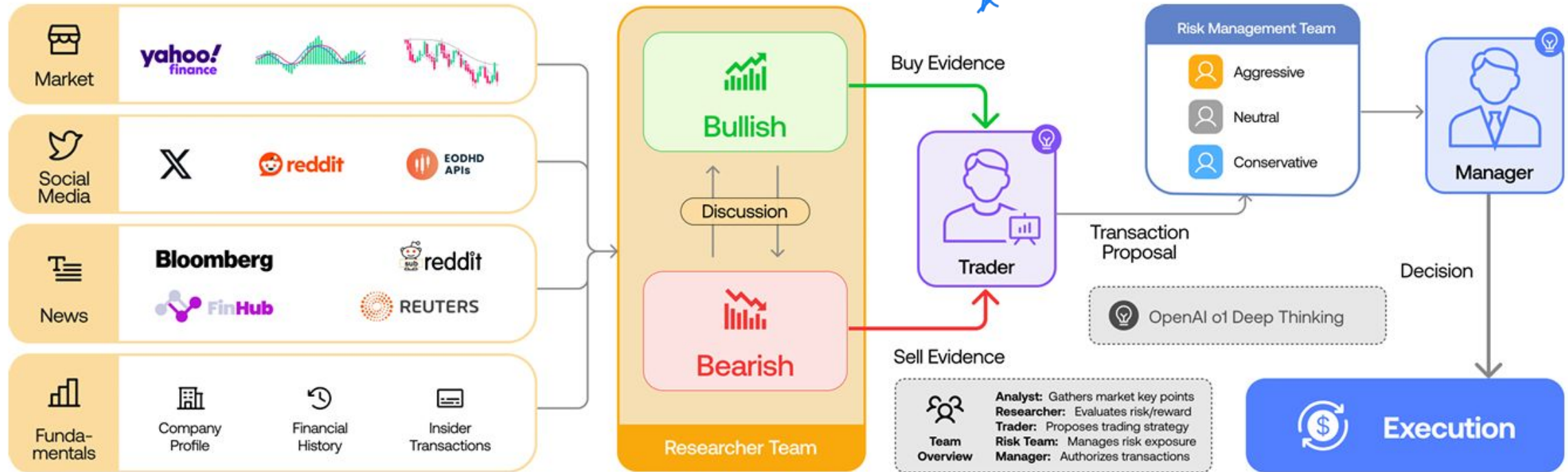


Fig. 1: TradingAgents Overall Framework Organization. **I. ANALYSTS TEAM:** Four analysts concurrently gather relevant market information. **II. RESEARCH TEAM:** The team discusses and evaluates the collected data. **III. TRADER:** Based on the researchers' analysis, the trader makes the trading decision. **IV. RISK MANAGEMENT TEAM:** Risk guardians assess the decision against current market conditions to mitigate risks. **V. FUND MANAGER:** The fund manager approves and executes the trade.

Experiments

- 期間: 2024/01/01 – 2024/03/29
- 作者使用自己整理的multi-asset, multi-modal dataset資料集
- 標的: 多檔科技股 (Apple、Google、AMZN)
- 決策限制: 代理只能用當天之前的資料, 避免讀去到未來資料造成偏差
- 輸出: 買/賣/持有訊號 → 執行交易 → 計算績效指標

Experiments

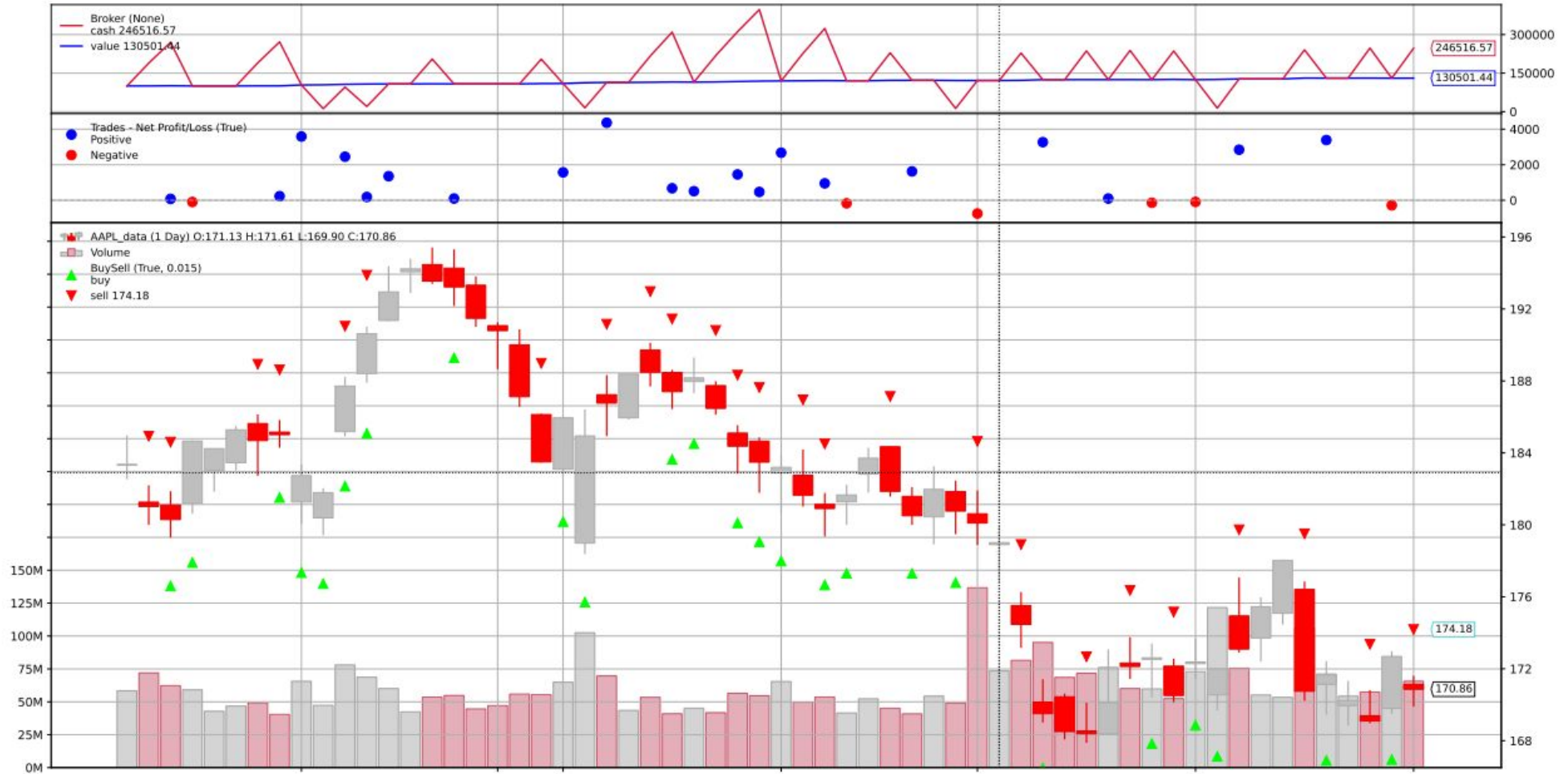


Fig. 6: TradingAgents Detailed Transaction History for \$AAPL. Green / Red Arrows indicate Long / Short Positions respectively, showing the model's trading decisions over time.

Experiments

Table 1: Performance comparison across all methods using four evaluation metrics. Results highlighted in **green** represent the best-performing statistic for each model. The improvement row illustrates TradingAgents’s performance gains over the top-performing baselines.

Categories	Models	AAPL				GOOGL				AMZN			
		CR%↑	ARR%↑	SR↑	MDD%↓	CR%↑	ARR%↑	SR↑	MDD%↓	CR%↑	ARR%↑	SR↑	MDD%↓
Market	B&H	-5.23	-5.09	-1.29	11.90	7.78	8.09	1.35	13.04	17.1	17.6	3.53	3.80
Rule-based	MACD	-1.49	-1.48	-0.81	4.53	6.20	6.26	2.31	1.22	-	-	-	-
	KDJ&RSI	2.05	2.07	1.64	1.09	0.4	0.4	0.02	1.58	-0.77	-0.76	-2.25	1.08
	ZMR	0.57	0.57	0.17	0.86	-0.58	0.58	2.12	2.34	-0.77	-0.77	-2.45	0.82
	SMA	-3.2	-2.97	-1.72	3.67	6.23	6.43	2.12	2.34	11.01	11.6	2.22	3.97
Ours	TradingAgents	26.62	30.5	8.21	0.91	24.36	27.58	6.39	1.69	23.21	24.90	5.60	2.11
Improvement(%)		24.57	28.43	6.57	-	16.58	19.49	4.26	-	6.10	7.30	2.07	-

Experiments

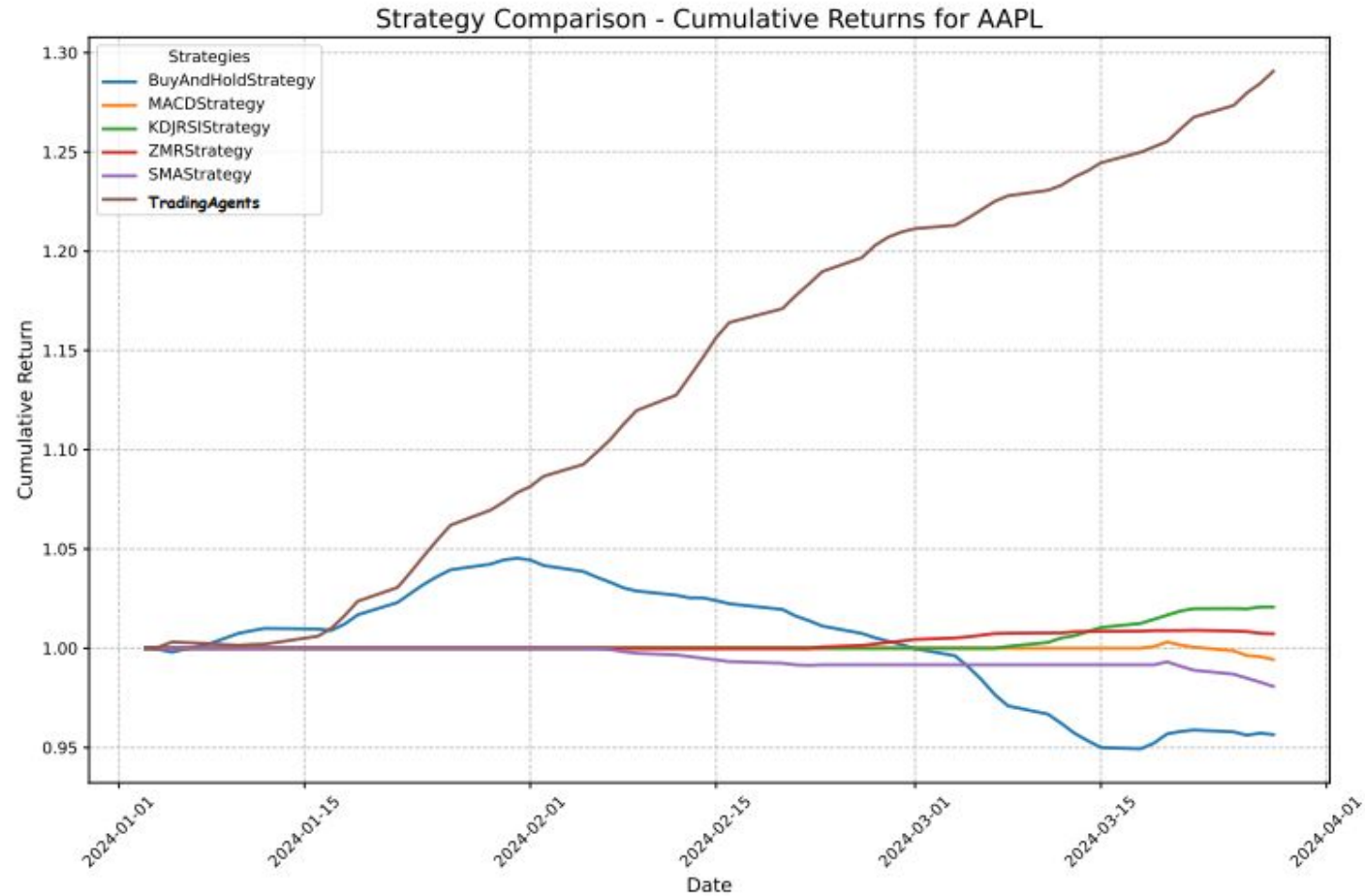


Fig. 7: Cumulative Returns on \$AAPL using **TradingAgents**. The figure shows the performance comparison of our model against baseline approaches for Apple Inc. stock analysis.

- 研究意義：展示了LLM 在金融交易中的潛力

<https://trade-bench.live/>

<https://nof1.ai/>

